Interactive Proof-Rule Guide

Tactix Development Project

August 8, 2026

Executive summary

Tactix now has a self-contained interactive guide to its proof-rule calculus. The guide makes the formal system easier to inspect without replacing the authoritative specification: every entry presents the calculus, a plain-language explanation, and two worked examples. It runs directly in a browser with no server or third-party dependency.

1 Findings and results

The guide covers all 22 public proof rules defined in `docs/calculus.md`, plus M4 (the backjump operation folded into M3) and V6a (the internal binary-resolution unit channel). The interface provides search, rule-family filters, collapsible cards, independent formal/informal/ example tabs, deep links, print styling, and a persistent light/dark theme.

The scope boundary is explicit. Rephase, SwitchMode, Walk, and Reorder are policy operations that change heuristic state rather than proof state, so they are not presented as proof rules. X5 Condition is also excluded because it is the catalog's documented no-implement decision; the guide links to that decision rather than inventing semantics for an unavailable rule.

2 Evidence

The deliverable is `docs/proof-rules.html`; the documentation map now links to it from `docs/README.md`. A structural audit counted 24 rule cards, 24 formal panels, 24 informal panels, 24 example panels, and 48 worked examples. HTML entity handling, repository whitespace, and the documentation size/progress checks were validated before landing. The formal content was checked against `docs/calculus.md`, with scope and implementation status cross-checked against `docs/implementation-notes.md`.

3 Risks and decisions

This is an educational derived view, not a second source of truth. The page states that `docs/calculus.md` wins if the two disagree. Future changes to rule semantics therefore need a corresponding guide update. Keeping the artifact as one dependency-free HTML file avoids a build system and makes it portable on the cluster, at the cost of a larger generated-looking source file.

4 Next actions

Open `docs/proof-rules.html` in a browser and use the family controls or search field to navigate. When a proof rule is added or its formal semantics change, update its card and examples in the same documentation change. Solver development otherwise resumes from the first action in `PROGRESS.md`.